



Map: Metropark guide, LH top: Table mat at local restaurant, LH Bottom: Oakland County Planning Commission Annual Report, RH Top: Detroit political weekly, RH Bottom: Report April 68 by Detroit Regional Transportation and Land Use Study.

THINK GRID

THE THIRD INSTALMENT IN THE CEDRIC PRICE SUPPLEMENT SERIES (see also AD 10/70 and 1/71)

Oakland Community College
Think Grid Michigan, USA

68

93

AC 35

31

3

Overall education thesis

To enable the maximum advantage to be taken of educational facilities, then such facilities must be sufficiently rich and varied in content to respond to the constantly changing demands made on them while, in form, location and duration, they must be readily available to the whole community at any time.

Our educational heritage is increasingly falling short, not only of the demands made on it but in its ability to be a generator of modes and methods for both the community and the individual to improve the condition of life. The "fall-short" is often due to the apparatus of education rather than its content or progenitors. More and more of us, not merely the disadvantaged, are learning in spite of education, not because of it.

Unless education is seen by all to be more than merely useful to the immediate situation, then the skill of living will owe less and less to education and thus education itself will become less and less socially important. Education and learning facilities must be produced primarily as a means of creating an essential part of the individual's need, like fresh air or pure water.

Thus, in this initial and brief investigation into the expanding role of Oakland County Community College, I have, as an architect and planner, investigated methods, techniques and artifacts whereby such aims can be more fully and more rapidly realised.

Comment

Emphasis on speed is important to Educational Bodies which, at their present rate appear to be doing fine. Gently, gently.



General architectural and planning thesis
It is felt that much of the reaction to change on the part of both "educator's and the educated" is due to the inherent inflexibility of existing and proposed buildings and physical communication networks.

The object of this investigation and the content of this report is also to outline those areas where I consider new radical decisions on aspects of future design, architecture and planning can be made to both aid and encourage the immense increase in the richness and comprehensivity of education and learning that will be essential to all communities in the immediate future.

US situation

Educational architecture/planning

Although immense advances are being made in existing forms of educational establishments and networks, I consider that increasing racial tension and school and college drop-out rates are just two of the many indications of a need for an entirely new form of educational servicing for the whole community throughout its life-time. It is not a question of supplanting existing modes of education but one of introducing an additional comprehensive service sensitive to needs scarcely yet formulated. The Community College would appear to be an ideal agency for such a task since fortunately not only has its role not crystallised into any particular authoritarian educational pattern, such as is the case with most universities, but also its very structure can be extremely sensitive to society's change.

There is every indication that Oakland County will be one of the most rapidly changing areas adjacent to Detroit City over the next 15-20 years. It will in fact become "a very large city in the sociological sense." (Oakland Planning Commission Report).

Comment

Oaklanders appeared to be extremely unaware of their own prodigious yet uneven physical and economic growth (see also following page)

ANNUAL REPORT
1966/1967

The Future:
Thinkbelt with Think-Links

△ Downtown Royal Oak, which with Pontiac is the tightest urban scene in Oakland Co.!

Typical new development at corner of mile square grid

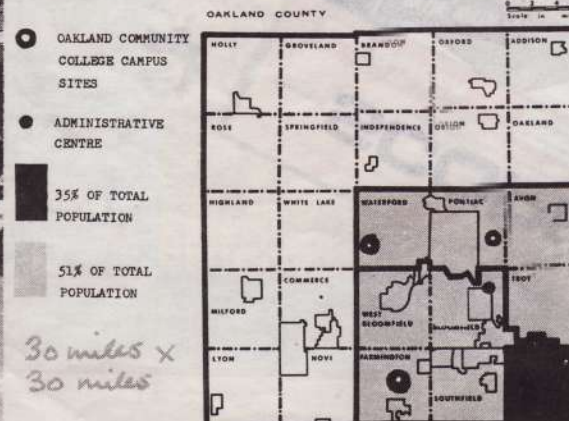
Commercial classroom

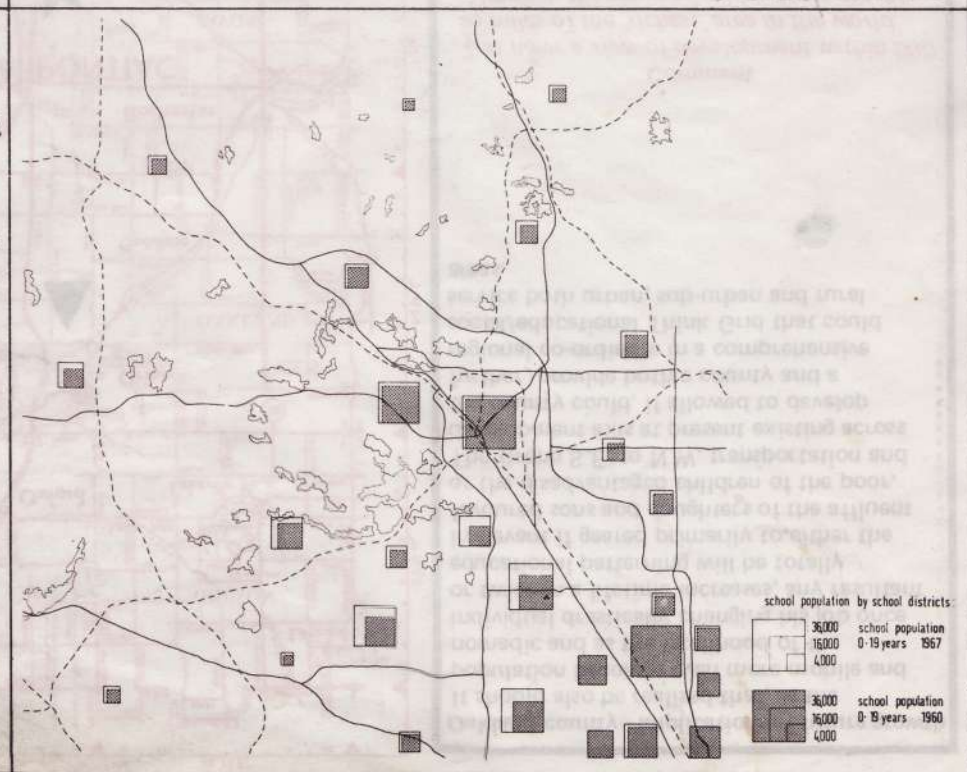
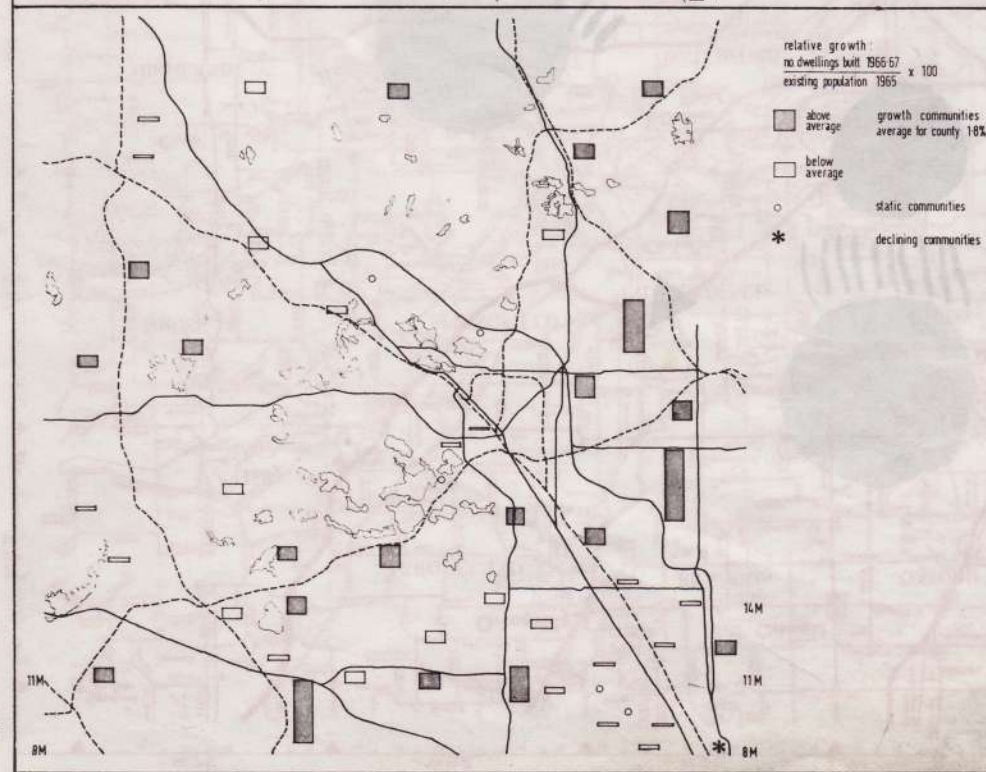
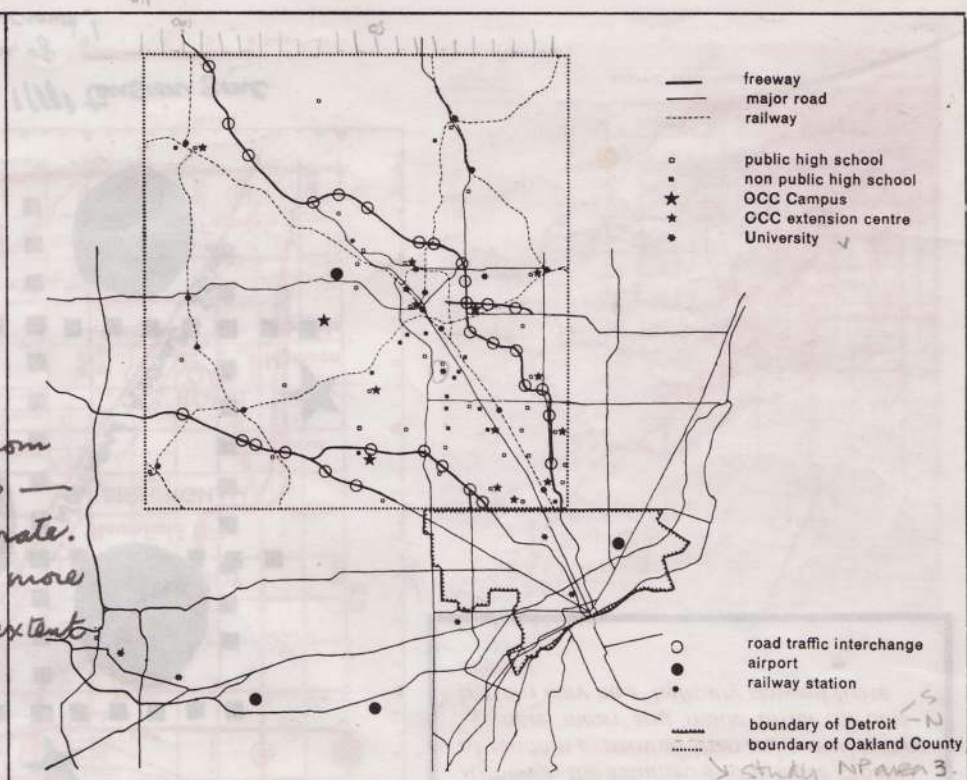
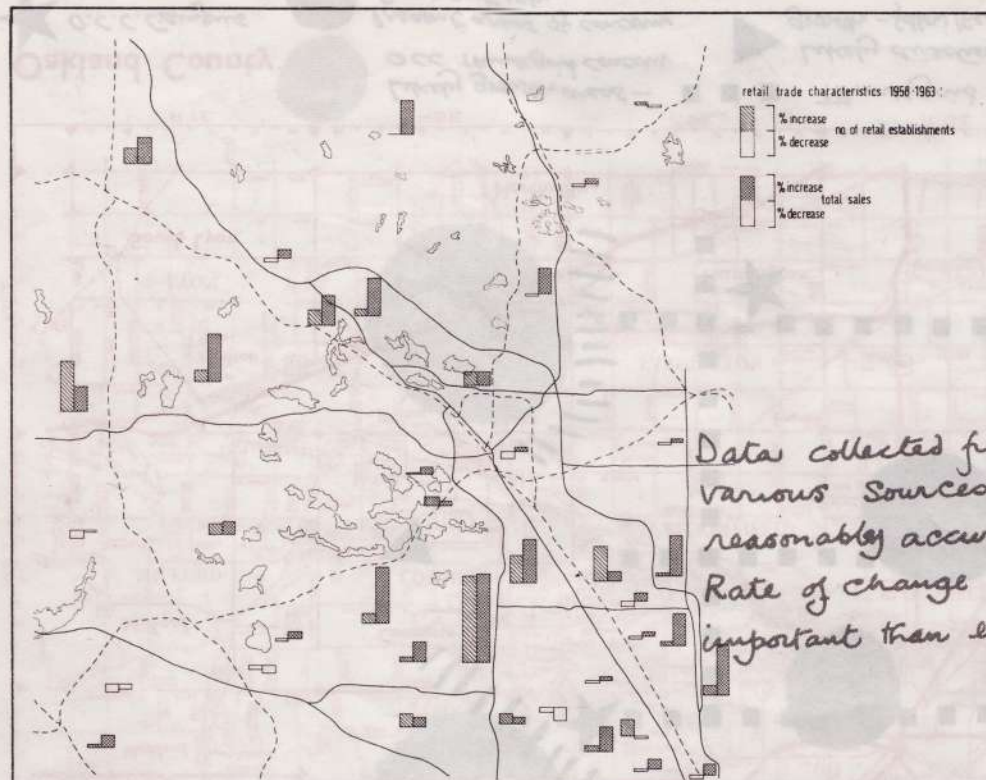
Florsheim shoes

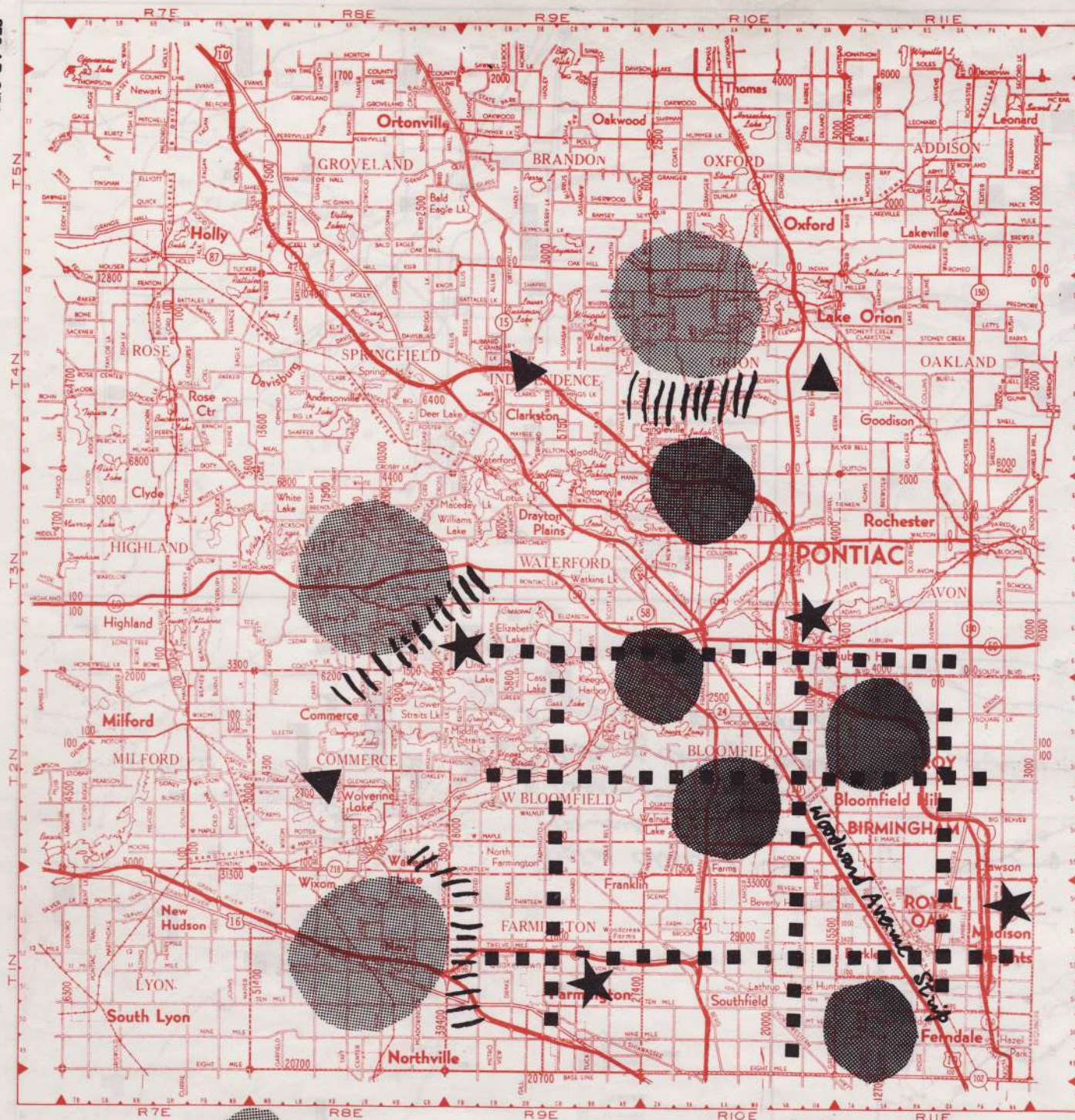
OAKLAND COMMUNITY COLLEGE



PLAN OF OAKLAND COUNTY WITH POPULATION & EDUCATIONAL LOCATIONS







Oakland County

★ O.C.C. Campus



Likely growth areas -
OCC Thinkgrid concern



Present areas of concern
Poor & Rich.



Thinkgrid !!!!! tension zone

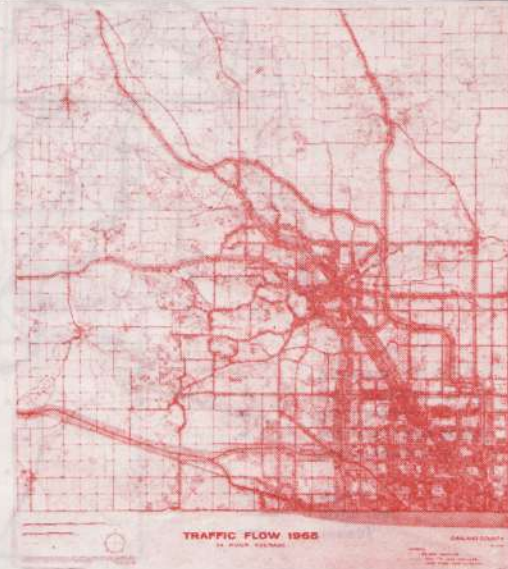
▲ Likely direction of
growth - follow the crowd!

0 1 2 3 4 Miles

Oakland county—implications of future growth
It should also be realised that, as the population becomes even more mobile and nomadic and as the likelihood of the individual drastically changing his job once or twice in a lifetime increases, any resultant educational patterning will be totally irrelevant if geared primarily to either the favoured sons and daughters of the affluent or the disadvantaged children of the poor. The strong S.E. to N.W. transportation and development axis at present existing across the county could, if allowed to develop further, provide both a county and a regional co-ordinate in a comprehensive social/educational Think Grid that could service both urban, sub-urban and rural areas.

Comment

Too naive a view of development within 900 sq miles of the 'richest' area in the world. Although the assumed grid is more capable of random expansion than say a semi-lattice — where 'mean' and 'mode' differ the best pattern may be a 'helically skewed finite drum'.



CRYSTAL LAKE - initial moves

1. Lakeside House

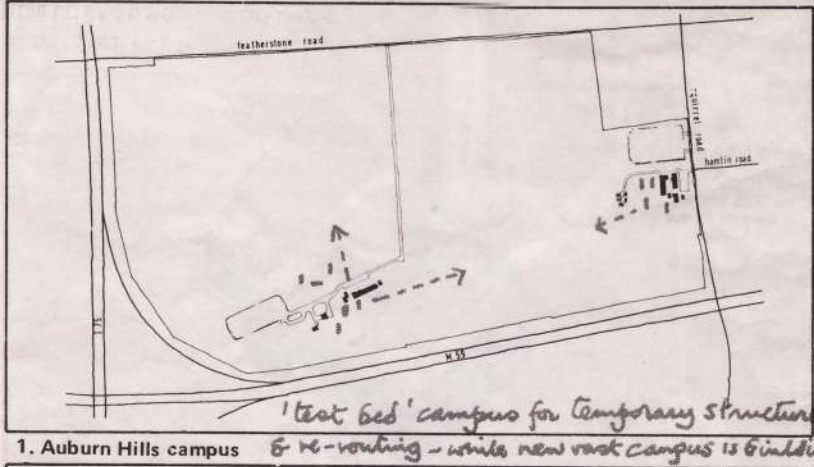
2. Hayes Jones Centre

FUN THINK STRIP

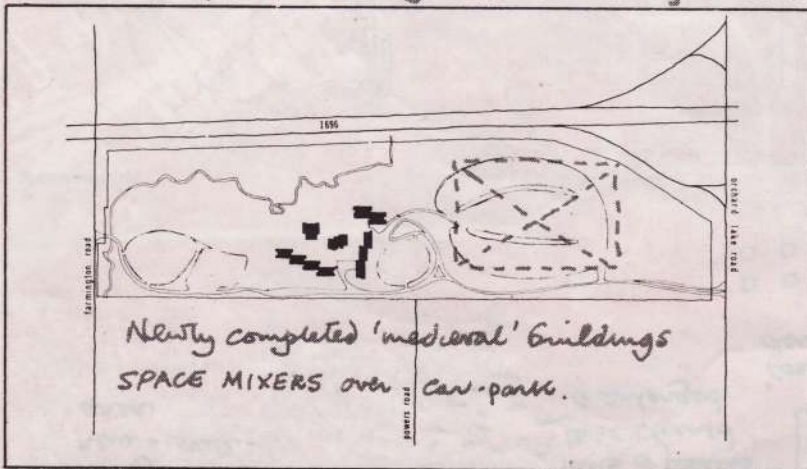
PROTOTYPE BRAIN FAIR LOCATION



Crystal Lake - a very poor black area - suffers from isolation through poverty which has nothing to do with distance. PHOTOGRAPH 9,000' WIDE



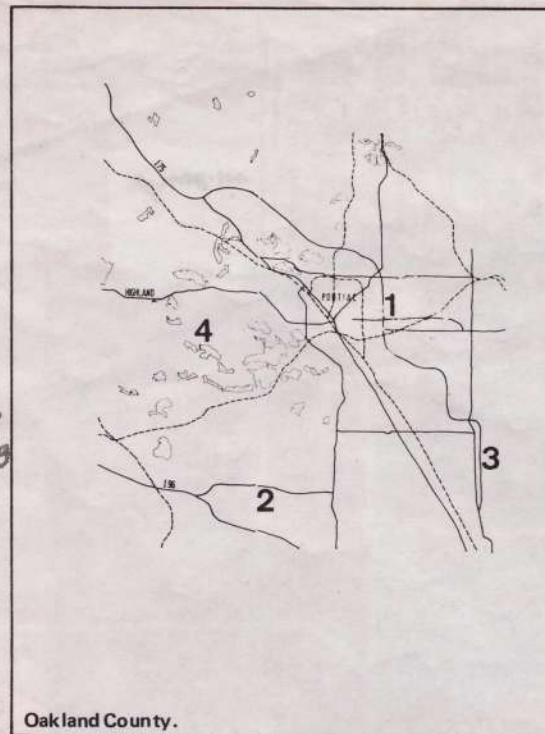
1. Auburn Hills campus



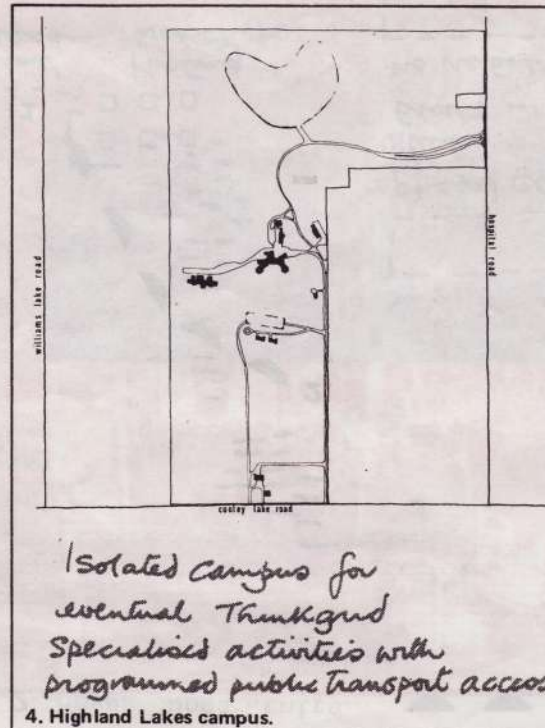
2. Orchard Ridge campus



3. Madison Heights campus



Oakland County.



4. Highland Lakes campus.

Proposals — general

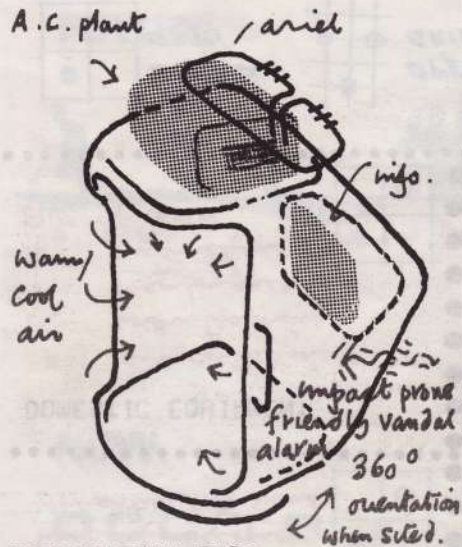
The existing and likely development pattern of Oakland County combined with the existing and potential capacity of O.C.C. make the establishment of a Think Grid a valid proposition. Such a grid varies from the original Thinkbelt* in so much that its capacity to service the area is multi-directional while any concentration of effort is located at any time, at a variety of points only a few of which would be permanent locations.

In addition, the adoption of a more even spread of the capacity to provide facilities should enable the boundary, and therefore the total area over which facilities may be provided, to be varied over a period of time. The following proposals are in 5 parts:

1. General procedure for implementation.
2. Expanded role of existing facilities, properties and programmes.
3. Elements required to establish a total educational/learning service — Think Grid.
4. Pilot projects.
5. Other related actions.

*See AD 10/66

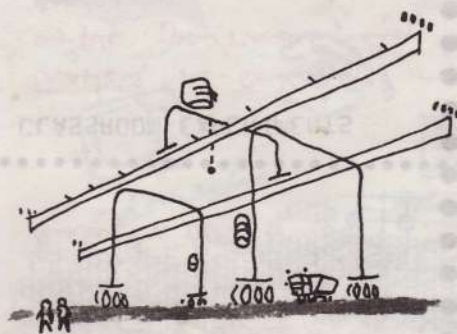
UNIT



INFORMATION PODS

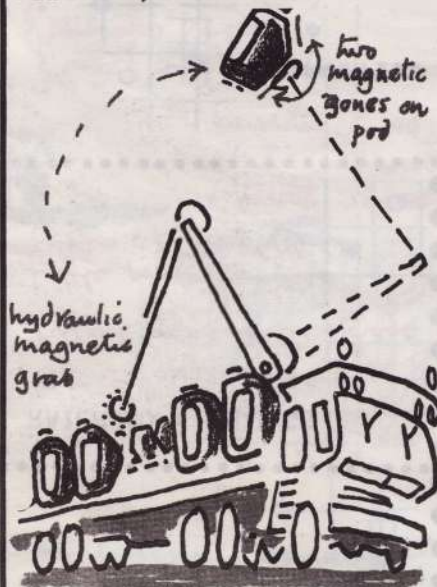
information link to SW/OCC
identification
Storage access comfort
Room for large mother + shopping +
two children who don't want to be there

OPERATION



SPACE MIXER
Screen screens
control shelter
tension + inflatable sky blind
air barriers
humidifiers coolers

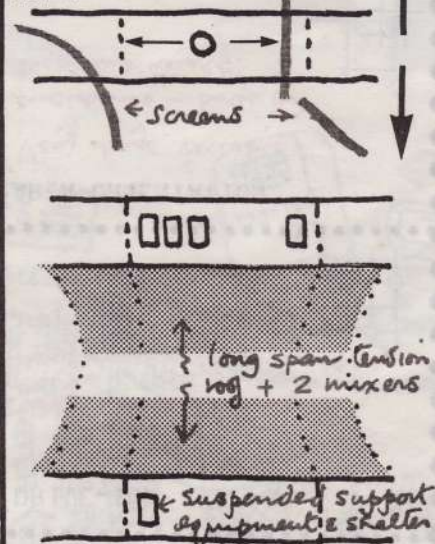
DELIVERY/COLLECTION



double stacked trucks
-pod batteries re-charged
en route.

OPERATIONAL

ALTERNATIVES



APPLICATION

24 hour cycle

Downtown
Royal Oak.



Proposals — particular

In effect, the first major educational programme suggested is that of explaining the relevance of O.C.C. to its own citizens. The existing campuses are at present reasonably well-placed in relation to that majority of the student population who have access to automobiles. However, the necessity of introducing a bus service is indication of the partial isolation of any fixed campus however apparently well placed on plan. Increasingly, the large fixed campus will become the place for educational specialisation for those with a capacity for maintaining a fixed timetable of attendance — not merely full-time students. However, in a mobile rapidly changing society with still a large number of poor and under-employed, the fixed campus cannot offer satisfactory shelter and servicing for all the activities O.C.C. should provide in order to justify its existence.

The primary requirement for the establishment of a Think Grid is the prior establishment of a total internal information, resources and programming network within O.C.C. sufficiently flexible to be able to respond to variation of demand.

Comment

'Once such a network has been established' — is the key phrase

PLANNING PROCEDURE

02-92

ONE TO FIVE YEARS

97-02

FIVE TO TEN YEARS

02+

OVER TEN YEARS

INFORMATION PODS

telephone, radio,
take away material
2-way T.V. later.



DRIVE-INS

Mass free-will time-up
with instantaneous
response mechanisms.



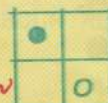
BRAIN FAIR

Public access to further
'private' scenes -
phased learners
attendance.
Information re-cycling
& re-reinforcement.



CLASSROOM EXPERIMENTS

trailers: re-servicing
un-fill: 'open' conditioning
gardening
auto + motor boat
handling
night-time projection



AREA ORIENTATION

Very large variable
enclosures - both in
area and height.



OCC INFORMATION SERVICE

Radio, T.V. station - Studios, field units
Telephone exchange - Booths
Personal information system
Specialised tutorial - campus.
Programming, record, storage
& retrieval - Agency deals.



STANDARD SOCIAL ADDITIVES

Swimming pools, beauty
parlours, house moving kits
workshops, etc -
through unexpected
mobility become
domestic extensions

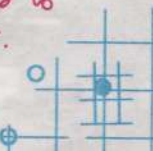


FLOATING WORKSHOPS

Domestic leisure
waterside time-up.

REGIONAL AND NATIONAL
OPERATION INVOLVEMENT

Co-operation invited to
encourage new O.C.C.
'clients'.
Rethink as continuous
planning feed.



DOMESTIC EQUIPMENT

e.g. air-conditioners
with teaching/
maintenance
instructions
& tools.

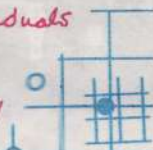
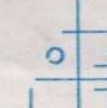
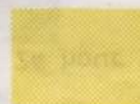


RAILROAD + SIDINGS

O.C.C. equipment &
freight movement + storage
Little public
access c.f. Thinkbelts

TRANSFERENCE OF
ACTIVITIES - REDUCTION
OF INFLUENCE

Capacity for individuals
to 'stand down'
Realistic initial
calculations on plant/
equipment
socio/economic decay.

CLOSED
GRIDOPEN
GRIDFRAGMENTED
GRIDTHRESHOLD OF
MAJOR PUBLIC
VISIBILITY

The difference between such a network and anything that may exist at the moment is primarily one of time - that is the time in which the system can recognise and respond to a variety of demands - such as result from racial unrest - which previously were pre-processed by other authorities prior to be considered an educational matter. Once such a network has been established then I consider the following additional elements are essential to its efficient operation.

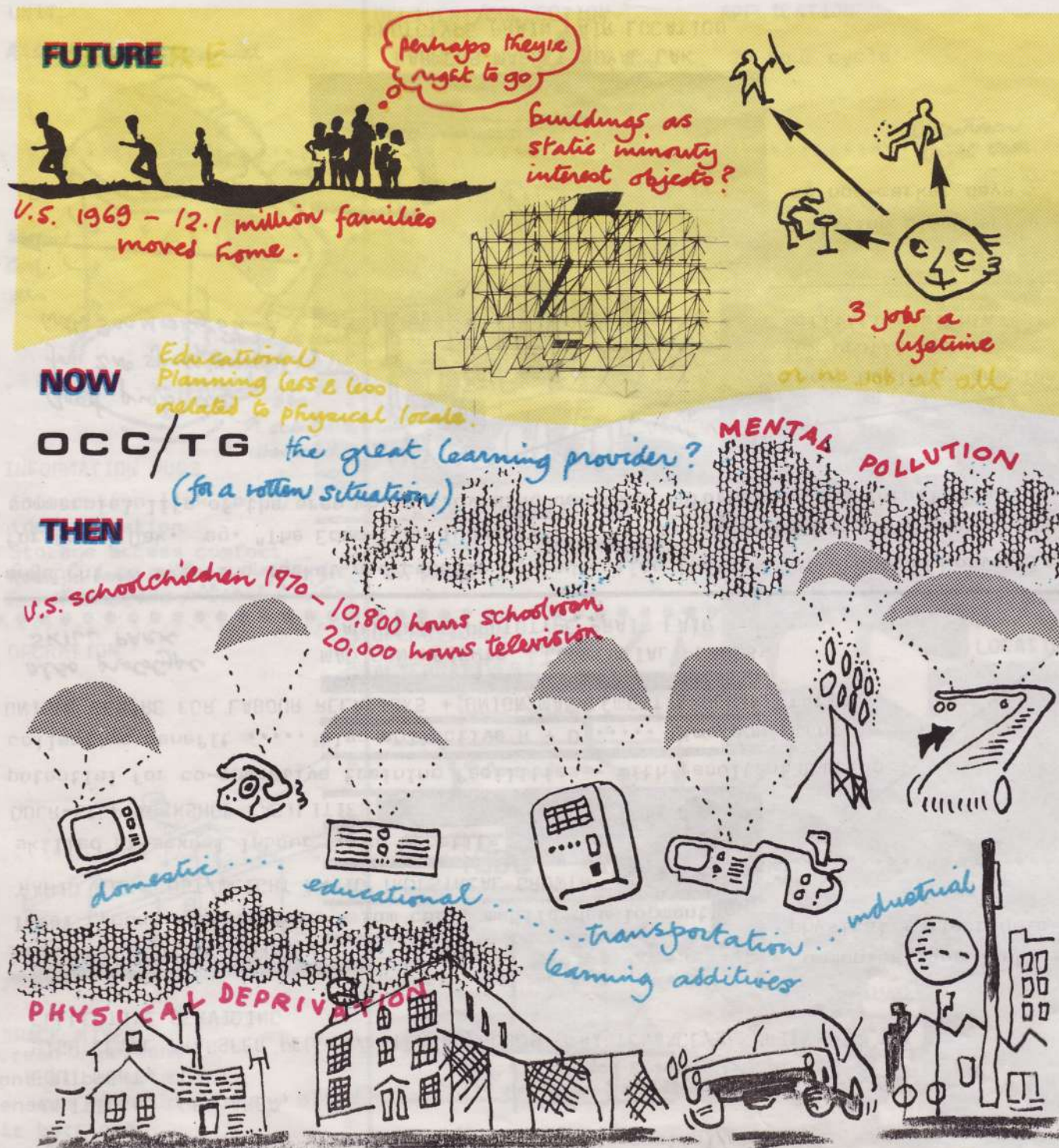
a. The manufacture and locating of not less than 100 O.C.C. information kiosks. Such kiosks would provide easy public access to information on O.C.C. activities and opportunities. Thus their form, and construction must enable rapid erection and removal.

b. The design and operation of an agreed method of communication between industry/commerce and advanced and adult education authorities including O.C.C.

c. The production of sufficient elements to produce at least 5 Brain Fairs at any one time together with agreement on at least 24 months of variable siting. These Brain Fairs, largely by means of highly serviced mobile elements, are able to provide rapidly an initial "appetizer" learning and information service as well as creating O.C.C. in microcosm.

d. The design, installation and operation of 10 major industrial/commercial "showcase" conditions - excluding any project at Madison Heights. These industrial/commercial showcases would vary in form depending on the particular industry. Primarily, they would help explain to the outsider the workings of the firm while providing for those interested further access to details of employment or, at least co-operation. While for the insider they would provide facilities whereby there could be two-way exchange between labour and management and O.C.C. faculty and students, thus producing the logical successor of the part-time professor.

e. The establishment of the Madison Heights Skill Centre as detailed. No upper limit should be set for numbers of "students" since in such a complex it is likely that there will be something approaching 24 hour usage with a large proportion of "overlapping" shift activity.



In determining the optimum design for the preceding elements — both systems and structures — O.C.C. should produce activity performance specifications for bids rather than the more traditional and far less acute "architects brief."

In all these new extended "teaching stations" it is likely that the Faculty should receive an additional "pioneer" salary allowance. Both Faculty and fittings must be first rate — and mobile.

Most pilot projects suggested are ones which in themselves can be used to test prototypes of the major elements of the proposals which are considered necessary for the total installation of a Think Grid.

a. The production and programming of a single brain fair.

b. The production of at least 12 O.C.C. kiosks. This number would necessarily be costly as they would not justify a "production run" — however, this is considered feasible in the service information they could provide both in their operation and eventual quantity manufacture. These should be placed in a parking lot, the entrance to a bad tenement, a prosperous suburb, a drive-in cinema, a slum street, a bar, a church hall, a factory exit, an old ill-equipped school, a shopping centre and an open-air recreation area. The acquisition or construction of rail sidings by O.C.C. should be investigated.

Related projects which, while not having the capacity for measurable response, cost effectiveness and social benefit, are considered worth instituting at an early stage include:

a. The utilization by O.C.C. of drive-in cinemas.

b. The availability of skill/techniques advice "on-site" at recreational and other open-air sites including Crystal Lake and Bloomfield

c. The tuning up of a particularly desperate area such as Crystal Lake or Royal Oak Township in which all forms of rapid humanization would have to be employed including kiosks, Brain Fairs, free air-conditioners, recreational instruction and equipment.

d. The utilization of the Grand Trunk Western Railroad to position kiosks, Brain Fairs and possibly special railroad trucks as detailed in the Thinkbelt proposals. The latter would form a separate performance specification programme.

Volumetric air-space - development potential - spatial/movement

ADDITIONAL WORK AREA,
EQUIPMENT, PLANT

HIGH LEVEL TRANSFER PEOPLE/MATERIAL/GOODS MAINTENANCE/SECURITY
COLLECTIVE SERVICING,

□□□□□□□□□□

level ground - short term - low cost, mobile development

RAPID HIGH COST/WEIGHT RATIO INDUSTRIAL GROWTH

skilled bi-sexual labour force appetite

DOOR-STEP WORKSHOP FACILITIES

potential for co-operative training facilities - with resultant sharing

collective benefit also collective R + D time/equipment

UNIQUE CENTRE FOR LABOUR RELATIONS + UNION/MANAGEMENT EXPERIMENTATION

also prototype
SKILL PARK

MADISON HEIGHTS . COMMERCIAL PLANTS
ADJOINING PROTOTYPE BRAIN FAIR

Communications links -
physical + electronic

adjacent to existing market and capable of indicating the theme of proposed Conventions for Royal Oak. eg. "The Education Industry". a central site taking advantage of the commercial life of the area while providing consumer information and intelligence

Good proto-site
for 2nd space max
with 300M sky blind.



Cheap building enables
low profit margin
activities within

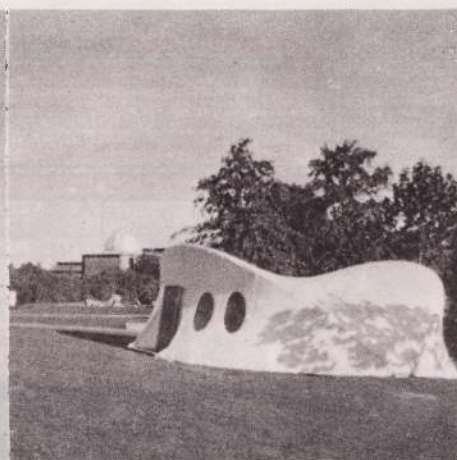
Existing slack space
on non-market days

Existing mobility

FARMERS MARKET ROYAL OAK
PROTOTYPE BRAIN FAIR LOCATION

b conta (Related proj.)

- Hills Golf Course. (Both, at present, areas of need.)
- c. The establishment of an O.C.C. radio station or, at least, time on other channels.
 - d. The insertion in poor housing of access not merely to O.C.C. but information in general. This may be little more than a free subscription to a magazine but should aim at providing free telephone access and workspace and, best of all, air-conditioning.
 - e. The introduction of a "leisure" industry into the lakes particularly where they occur adjacent to depressed areas e.g. Pontiac. O.C.C. can provide information and instruction not only for the users of the lakes but also to the possible dispensers of lakeside services. (c.f. T.V.A. and Merimac Project 1962-63)
 - f. Possibility of floating classrooms on lakes for all forms of instruction and information from life saving to power-boat technology.
 - g. Installation of large O.C.C. information screens on major transportation routes. These should not merely advertise O.C.C. but show - as on air-line signs - exactly what is happening and where. They should be programmed from the information centre.
 - h. A field study of the Woodward Avenue "strip" to see which forms would co-operate with O.C.C. in advertising and providing kiosk or screen space.
 - i. The assessment of what information is being constantly collected by such bodies as The Oakland Planning Commission, TALUS etc., that could be useful to O.C.C.'s is forward planning and the establishment of a method of continuous information feed from such bodies.
 - j. A wide-cast look at the possibility of O.C.C. sharing space, information, mobility or energy facilities with such bodies as the police, transportation authorities, health authorities, energy producers, newspapers, etc.
 - k. A coherent method of giving information and advice to such groups and authorities as Wayne County Education Authority, the education committee of the New Detroit Committee (New Detroit Incorporated) and anyone else in community need.
 - l. An assessment of the educational/learning value of such standard products as: a fully furnished mobile home (housecraft etc.) an air-conditioner (electronics etc.)



flown from California in airline container .. trucked to site .. erected in 3 hours .. removed in 1 hour - the Thinkgrid exposition lounge with Cranbrook planetarium

■ ■ ■ ■ ■ A CLUE TO TUNING UP THE RICH 'DISADVANTAGED' AREAS ? ■ ■ ■ ■ ■

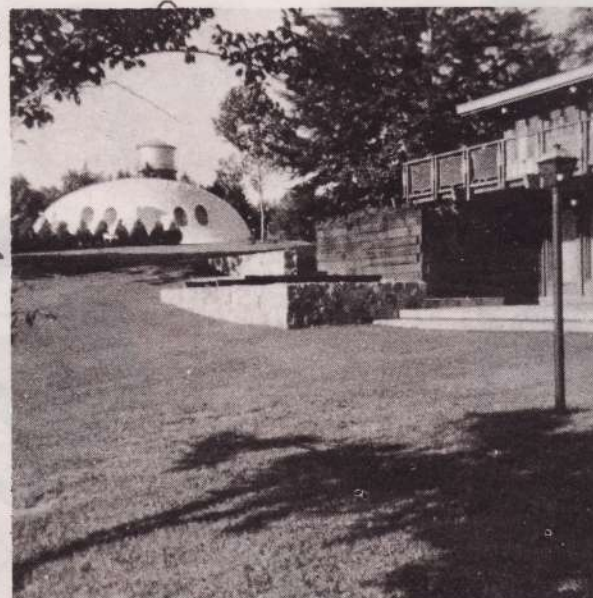
maximum use of semi-controlled open environments for testing alternative futures

Planetaria only a start .. meteorological measurements, radar, pollution counts, artificial rainmaking etc.

WARM AIR CURTAINS

CONTROLLED BALLOON LIFTS

ULTRA-VIOLET LIGHT ZONES



'Spectator' information
domestic internal
information/learning

CURRENT AUDIO-VISUAL TOYS

GARDEN/RURAL TOYS/TOOLS
paced-phased nature walks
garden/rural experience
and expertise
MOTORISED LAWN MOWER
SAFETY *ref. David Greene*
HORTICULTURAL MAGIC KITS

movement potential of
community facilities

movement potential of
land use previously
related to ownership
eg garden parties

AIR CONDITIONED HALL
£3,000-00

AIR CONDITIONED HOUSE
£250,000-00

m. Study of possibility of single pilot projects such as:
Jobmobile (Norwalk, Conn.)
Mobile Swimming Pool (Detroit)
Mobile Workshop (Det. Edison, G.M. and others)
Mobile beauty parlour and hairdresser (Greater London Council U.K.)
n. Design investigation, with possible invitation to industry to bid on particular performance, specification, into various forms of cheap, rapidly adjustable total shelter for large areas (min. 2000 x 2000 feet) enabling rapid conditioning of ordinary activities for further educational/learning opportunities.
o. O.C.C. should develop a capacity to accommodate what was previously considered a disaster or panic move. This means there must be an educational/learning emergency pack available but without any "hand-down" connotations. (e.g. The Carver "Freedom School" movement).
p. Easier, more effective methods should be pioneered by O.C.C. in relating Oakland County citizens to their physical environment. (e.g. Controlled balloon lifts for O.C.C. intake).

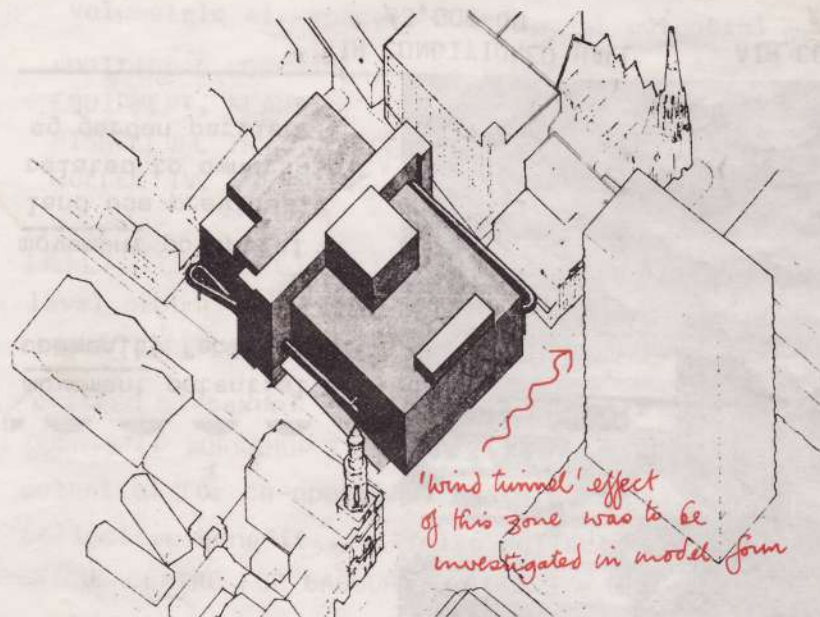
Recommended schedule for proposals

Initial research	Year 0 - Year 1
Information Centre	Year 0 - Year 1
Pilot projects	Year 0 - Year 2
Total Grid	Year 2 - Year 4 or 5

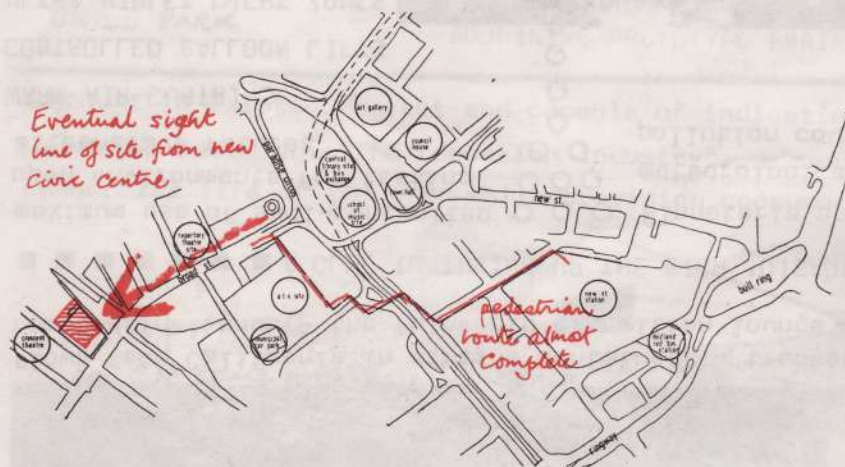
5-10
10+

Comment

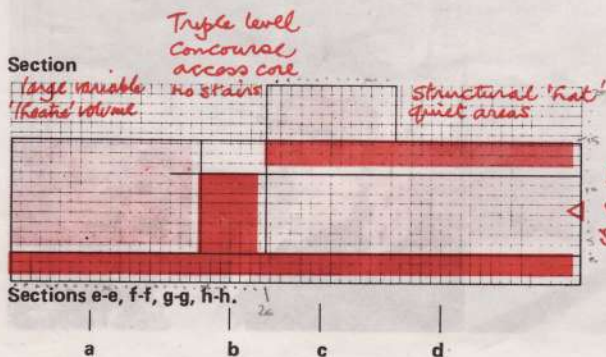
The compartmentalisation of community learning servicing - even in 900 sq miles lumps - is at initial planning stage too limited. We've all seen most of the hardware before, I still think the siting good but rather too 'sensitive'. 'Invisible servicing' is the most valid part of these proposals and such servicing could well produce new architectural and planning priorities.



Axonometric

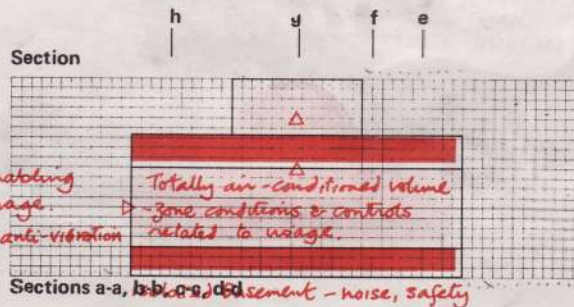


Site plan.



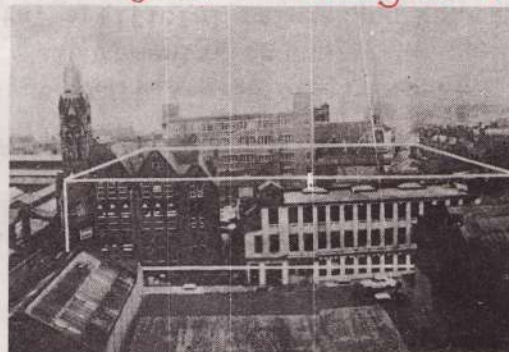
Major volume enabling varied volume usage.

Suspended floors for anti-vibration



↑ looking west

looking east ↓



69

94

A13

32

2

BMI

HQ/Birmingham and Midland Institute,
Birmingham, UK

Initially the BMI commissioned an independent investigation of their own possible futures. It was decided that the future of the BMI lay in radical change and expansion. The brief included further investigation of the feasibility of a new building or buildings and an area communication network.

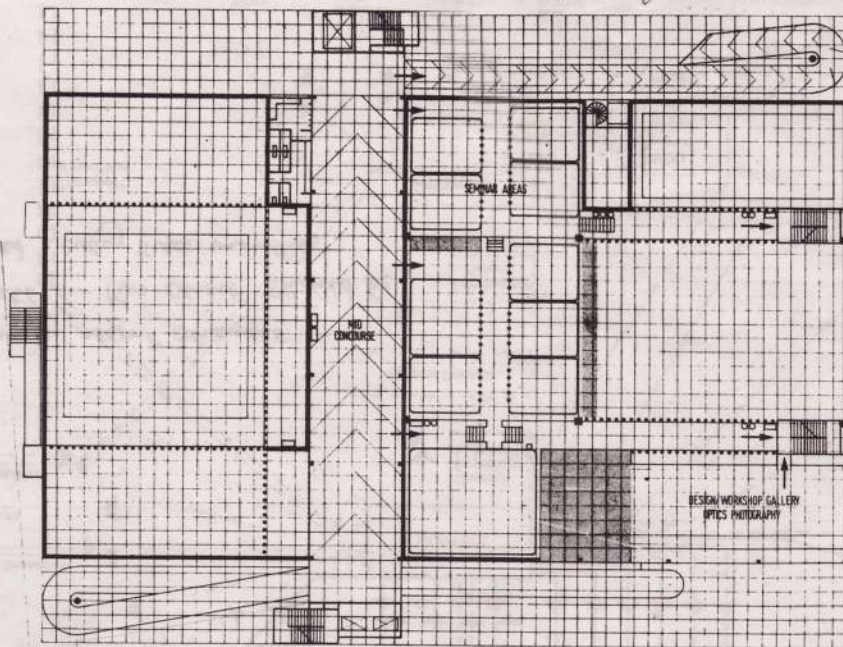
Extract from initial design submission to client:-

Why a new BMI/HQ?

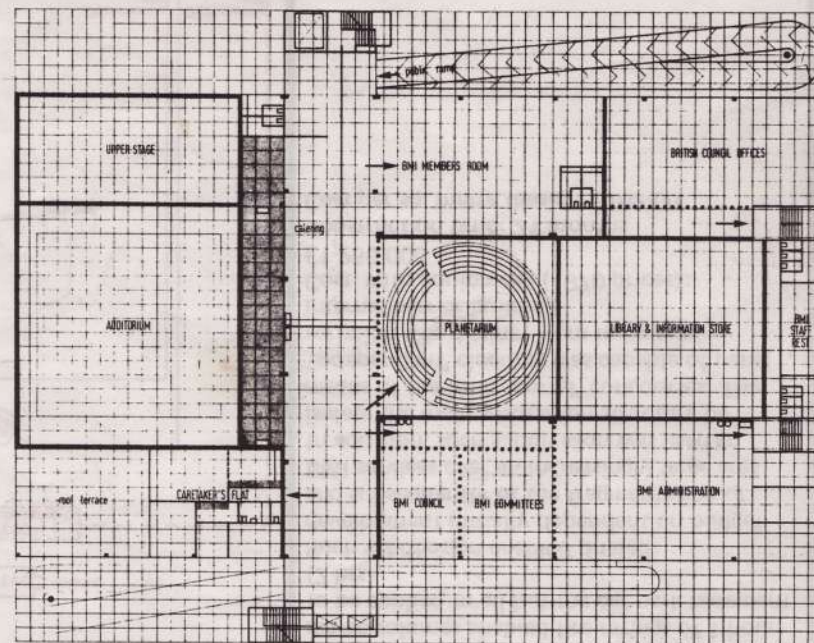
The increase in appetites and opportunities for self pace learning for all ages is likely, in the near future, to generate the largest single social industry. No activity building, whether library, theatre or foundry can meet the variety and frequency of likely demand. Indeed, with the growth of learning facilities in every home, no public building need in itself offer all the facilities required. However, such a building as that proposed can provide the continuous scientific, technological, cultural 'servicing' required by both the individual and the group. At the same time the building can act as the nerve centre, producer and distributor of an enormous range of services to members and the public in the region.

The Birmingham and Midland Institute, set up in 1854, concentrates on surveying the whole field of science, literature and the arts with the intention of directing efforts and finance onto areas where little or nothing is currently available to the general public.

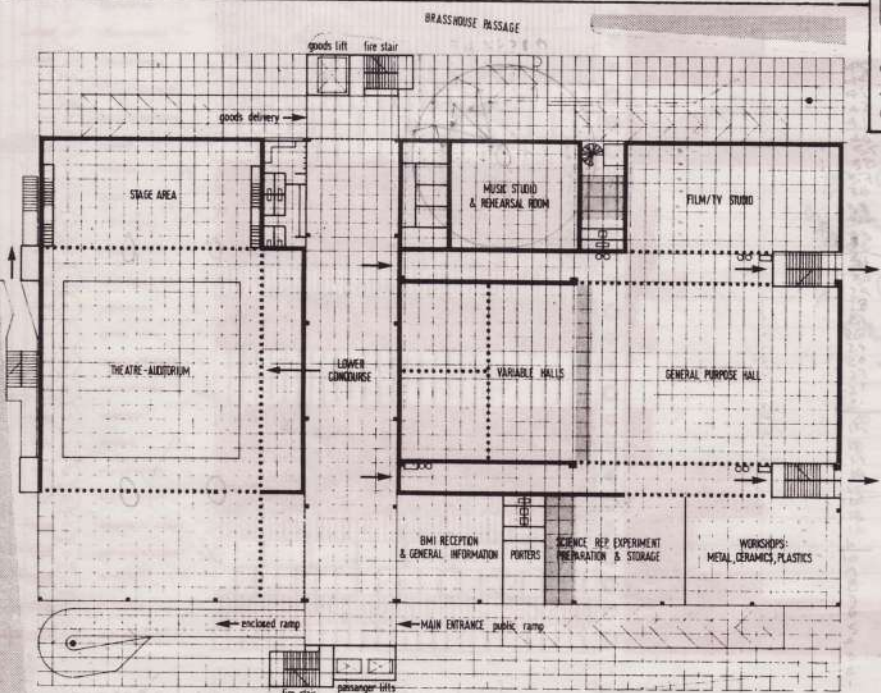
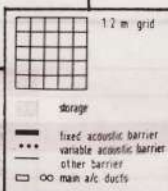
It's a bit sad that the most modern theatre now still require conventional flexibility - rather than a new scene - SEE 'THEATRE' SIDE OF BUILDING



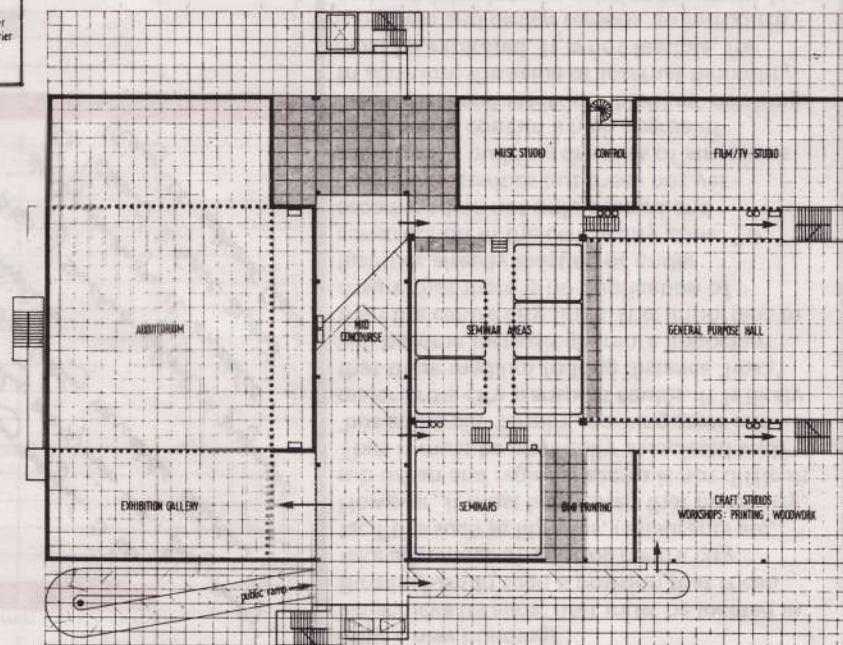
Plan: upper grade (at datum + 9.6)



Plan: intermediate grade (at datum + 5.6)



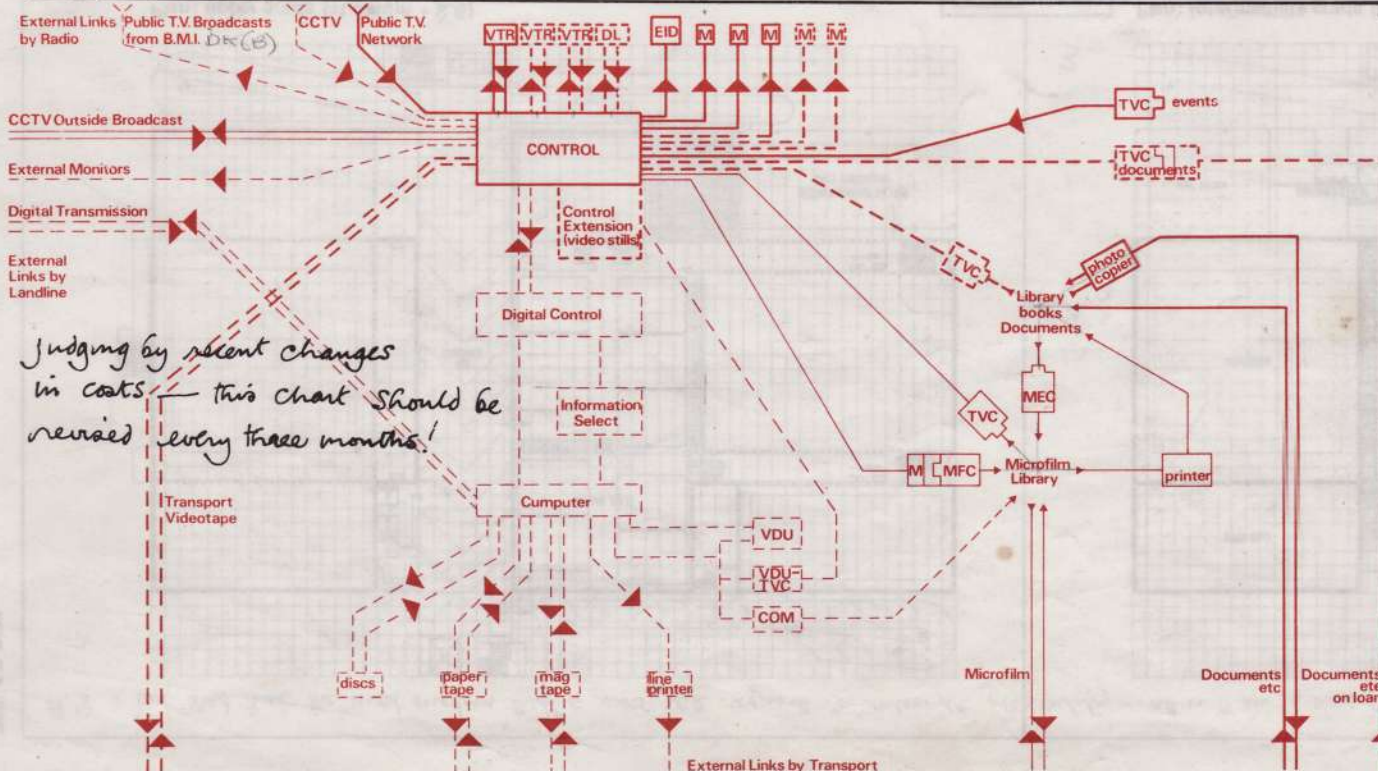
Plan: main grade (datum)



Plan: intermediate grade (at datum + 4.35)



Activity / area / location / frequency / capacity Chart



Staged networks

Why a new building?

To achieve its aims, BMI must be located in Birmingham in such a position that it can take advantage of existing centre city facilities such as libraries, restaurants etc. However, it must be so sited that outer city, sub-urban and regional access is not difficult. Thus it must be physically associated with 'bus and train' termini, parking areas and major cross-city transport routes. It is likely that these medium to long distance 'links' will become increasingly more important than the centre city links. It is important to ensure that there is no duplication of facilities already existing or under construction in the centre city zone. To accommodate not only present and projected activities but those not yet envisaged, a building or series of structures will have to contain a high degree of mechanical and electrical 'blanket' servicing. In addition, the value of internal flexibility must be exploited. It is not feasible, for economic and design reasons, to introduce such servicing into an existing structure. It is therefore considered that on the grounds of both siting and inbuilt facilities that a new building is required.

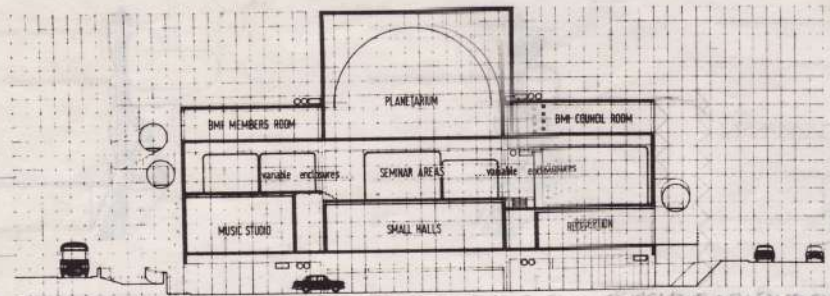
The sub-dividing of the building into five major volumes enabled fire ratings of 1 hour to be applied to all but the car park basement (2 hrs).

Such subdivision enabled phased development initially dependent on budget and subsequently of the fragmentation of BMI services over an increasingly large area. In addition, such subdivision was used to isolate the main suspension structure by means of the infill 'portal' concourses. The 'independent' cladding was capable of panel variation was distinct from individual internal barrier building.

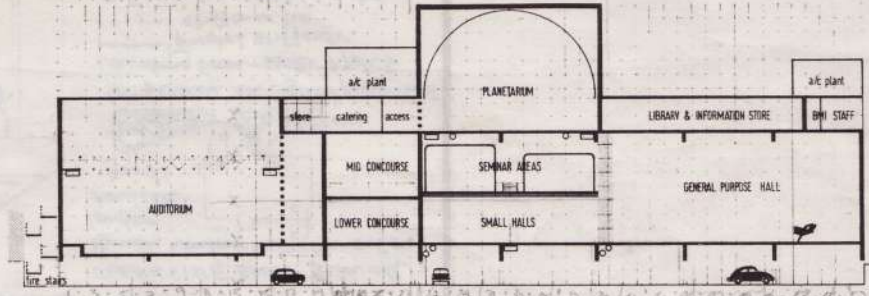
Cost, with equipment (1970 prices):
£1,250,000

Construction time: 21 months

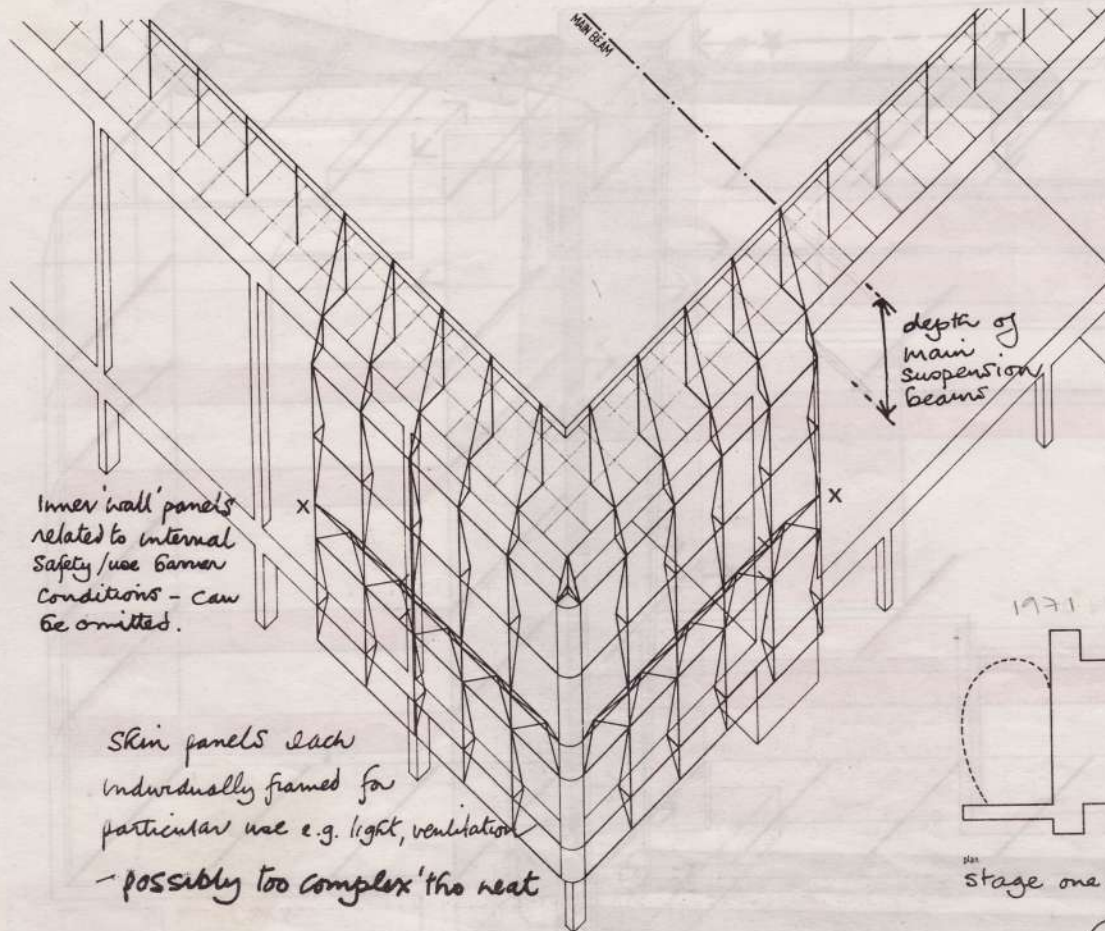
Cost per sq. metre (basic): £102.00



Section c-c



Section g-g



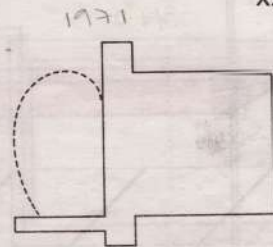
Inner wall panels related to internal safety/use barrier conditions - can be omitted.

Skin panels each individually framed for particular use e.g. light, ventilation - possibly too complex tho neat

XX Elevation

hinged & braced 'sliding' skin support framework

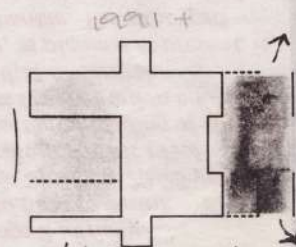
XX Plan



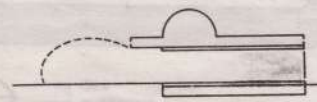
stage one - for economy!



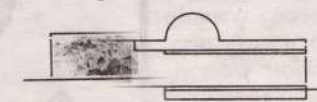
stage two



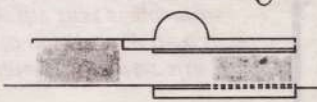
totally enclosed volumes replaced by shelter for mobile services in final stage 20+ years

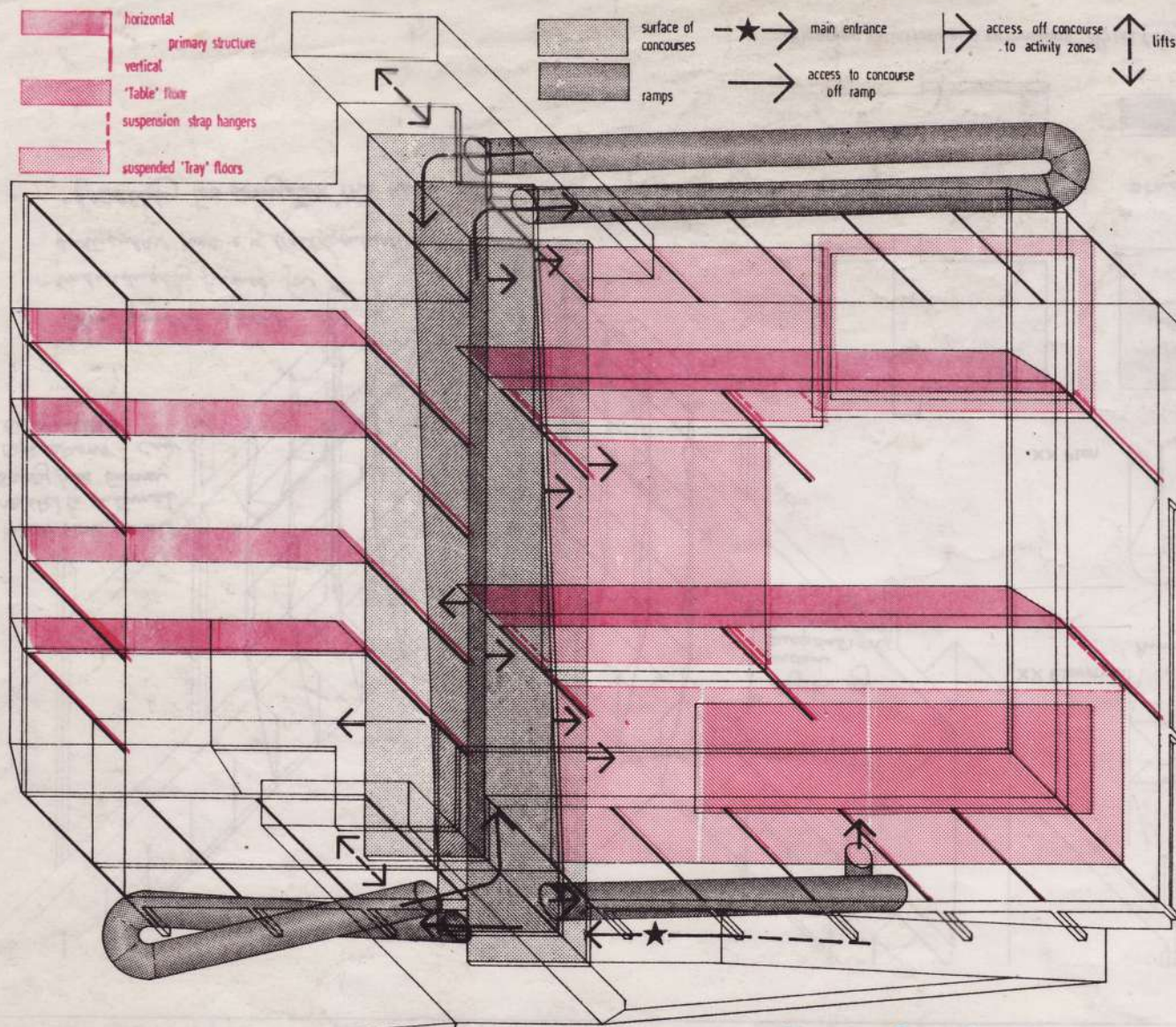


Planning variations and extensions



Buildings can 'shrink' in order to grow.

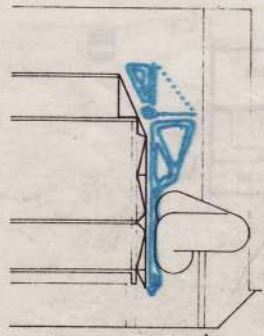


**Comment**

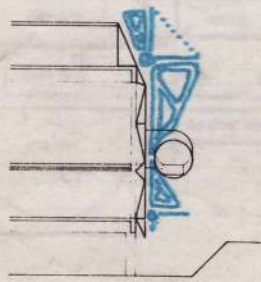
All work stopped late 1970, after site investigation, due to new policy by a new Council. Previously the first stage, with inflatable 'temporary' theatre had been abandoned as a result of a 'cost-over-time' assessment of the advantages of building the more expensive Stage 2 initially. This building would have to shrink physically (see stage 3) for its 20-25 year usefulness to be exploited. This I feel permits the heavy activital loading of the site initially. I hoped its fragmentation could be co-ordinated with that of central Birmingham which, as planned at present, is almost too hot to handle. The proposed communications network may be applied elsewhere — as is the cleansing boom.

Structure systems

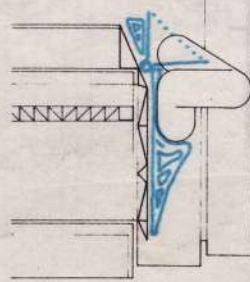
In effect, 2 systems — one portal — one suspension with floor-deep hanger deck — sliding in-fil concourses.



Sections thro' skin a-a

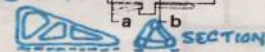


b-b



c-c

CLEANSING BOOM. 3 in all. Human window/wall cleaners no longer available — good!



SECTION
low pressure air-inflated pyramidal
cleansing boom — water + polish
— pivoted stiffener
..... stiffener tie
○ horizontal crawling motor.